



Commercial Aspects of Oil Refining

Refinery Economics · Crude Selection · Product
Trading · Cargo Operations



Commercial Aspects of Oil Refining

Valuing Crude Grades, Refining Margins and Benchmark Pricing in Global Markets

Commercial Aspects of Oil Refining is a 3-day course that explains the physical and commercial drivers of refinery operations and economics. It gives participants practical knowledge of the key parameters that determine refinery profitability—refining configurations, feedstock costs, product yields, margins, crude supply choices, product trading, marine freight and project economics.

The programme shows how crude supply and products trading can be used to improve refining margins, and how commercial decisions are made in different business units to optimise the refining business in a competitive market. Using real-life examples from refining and marketing organisations, the course bridges theory and practice so participants understand how refineries are commercially operated, how market structures and prices affect margins, and how trading tools and operating parameters can be used to enhance profitability. Morning sessions focus on the core concepts behind refining economics; afternoon sessions apply them through case studies and problem-solving exercises.

 LOCATION

Rio de Janeiro

London

Singapore



COURSE OBJECTIVES

By the end of this course, participants will be able to:

- Explain the fundamental principles of refinery operations and economics, and the main cost and revenue drivers in a refining business.
- Describe different refinery configurations and assess how configuration choices impact economics, product yields and margin potential.
- Calculate feedstock costs and refining margins, including GPW and other key margin metrics, for different crudes and operating modes.
- Understand how crude oil and product markets are structured and how market conditions influence refining and trading decisions.
- Identify ways to optimise refining operations and align throughput, product slate and crude selection with commercial objectives.
- Calculate freight costs and voyage economics, and assess their impact on supply routes, netbacks and trading margins.
- Analyse arbitrage opportunities and calculate trading margins for crude and products, including use of swaps and backhaul economics where relevant.
- Negotiate and review oil sales/purchase contracts with better understanding of the commercial terms and their implications for refinery and trading profitability.
- Manage cargo operations and documentation from a commercial perspective, including delivery modes such as FOB, C&F and Ex-Ship.
- Develop and operate basic business plans for refining and trading activities, integrating operations, pricing, logistics and risk management.



This course is designed for professionals involved in downstream oil refining, supply, trading and commercial decision-making who want a clearer understanding of how refineries make money and how trading and logistics influence margins. It is particularly relevant for:

- Oil and gas engineers
- Planning engineers and managers
- Marketers and traders
- New entrants to downstream oil functions
- Business analysts
- Product managers
- Banking and government officials dealing with petroleum downstream
- Finance and accounting analysts
- Procurement and sourcing professionals
- Investors and stakeholders evaluating refining assets

Overview of Refining Process and Operations

- Introduction to refining process
- Overview of refinery operations
- Refining operations
- Refinery configuration
- Impact of configuration on economics
- Refinery operating cost
- Crude oil types and characteristics
- Crude selection and types of feedstock
- Products yield and GPW
- Calculation of refining margins

Case Study

Assessing three different refinery configurations and their impact on refinery profitability.

The attendees will learn which configuration and mode of operation will yield the highest refining margins while selecting the best crude and feedstock.

Problem Solving

Calculating gross refining margins for different crudes for base load and incremental crude runs and identifying optimal mode of operations. The analyses identify whether to make or buy any specific product to meet products' supply commitments.



Refining Economics & Planning

- Introduction to refined products
- Products grade and classification
- Products specification and its implication
- Products blending (gasoline, gas oil and fuel oil)
- Gross, variable and cash margins 3-2-1 crack speed
- Refinery economics and planning
- Developing operating and business plans
- Project economics and cash flow
- Refinery investments

Case Study

Developing optimal product mix and crude throughput levels for a typical refinery from a range of crude oil grades which will maximise profits under different pricing scenarios. The attendees will also learn how to evaluate refinery projects and cash flow projections from refinery investments.

Problem Solving

1. Calculate 3-2-1 crack spread under different crude and pricing environments and making commercial decisions for selecting crudes and refining operating modes
2. Running simulations for products blending, including octane/economics for gasoline and cutter stock for fuel oil.

Attendees will also learn how to minimise economic penalty for quality giveaways.



Global Refining, Supply and Trading

- Overview of global refining
- Physical oil markets
- Market structure and regional trading hubs
- Crude oil and products' pricing
- International oil supply and trading
- Cargo delivery modes (FOB, C&F, Ex-Ship)
- Products arbitrage trading
- Shipping and freight economics
- Cargo nominations and operations
- Price exposure and risk management
- Principles of hedging
- Oil contracts and its implications of commercial terms

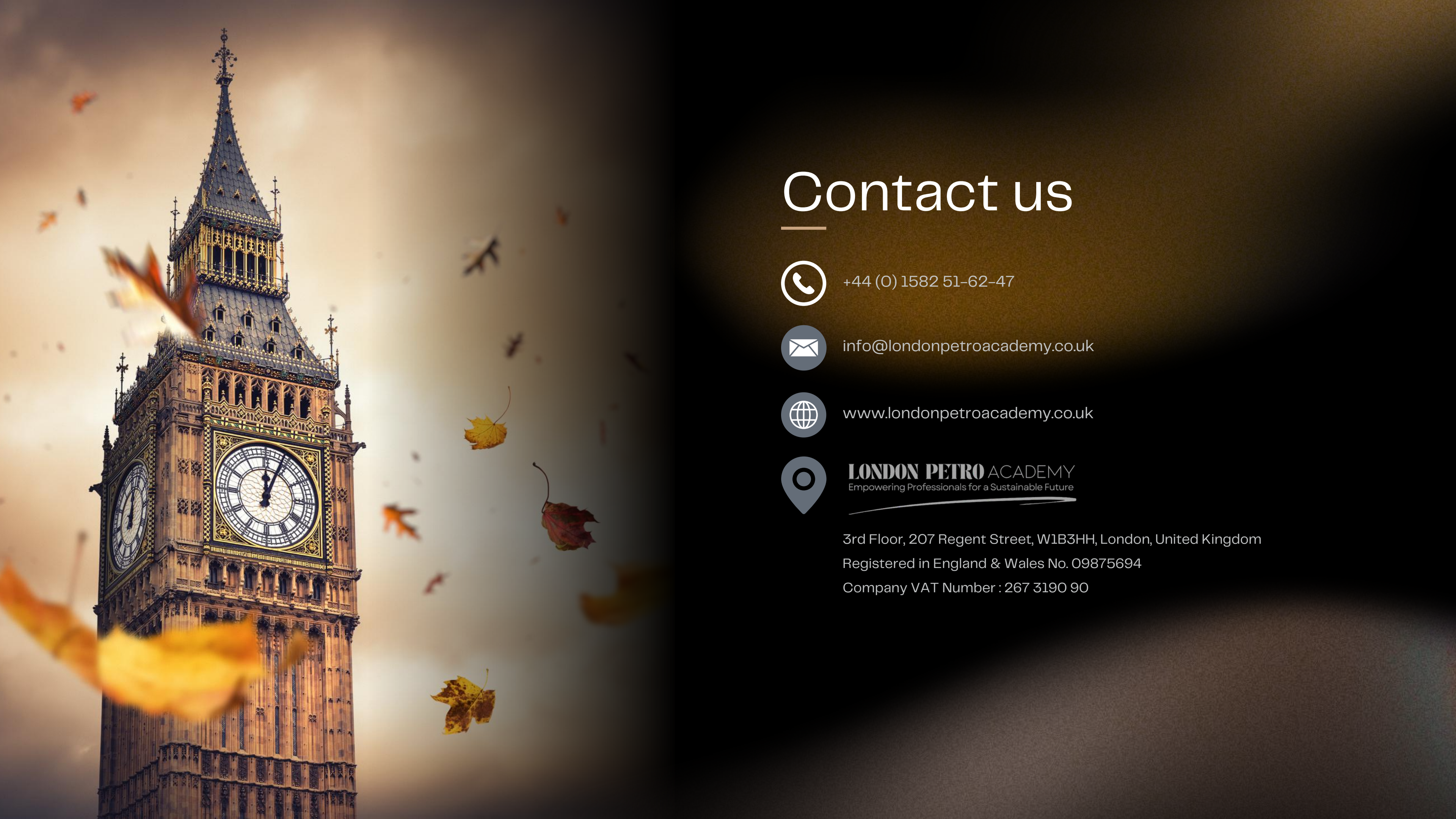
Case study

Running an interactive simulation for crude oil and products trading involving different pricing structures and various delivery modes in a real world. The attendees will also learn how spot barrels are traded and how arbitrage trades are done. The second case study will focus on shipping economics and cargo operations and how economies of scale in shipping can affect trading margins.

Problem Solving

1. Market structure and its implications on storage economics
2. Calculation of netback pricing
3. World scale and voyage costing
4. Managing trading risk using future contracts and hedging techniques
5. Analyzing oil contracts from commercial aspects





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